⇒ Principal of Complete Induction

Let $p(n)$ be a statement defined on the integers $n \in N$ such that

i) $p(m)$ is true for some $m \in N$

2) Whenever $p(m), p(m+1) \ldots p(k)$ are true then $p(k+1)$ is true, where $k \geq m$

Example:→ prove that any integer $n \geq 2$ is either a prime or product of prime no.

Soln:→ Let $p(n)$ be a statement that n is prime or product of prime

i) $p(n=2)$, $p(2) = 2 \times 1 \Rightarrow$ prime

$p(3) = 3 \times 1 \Rightarrow$ prime

$p(4) = 2 \times 2 \Rightarrow$ product of prime

ii) Let assume $p(n)$ is true for $2 \leq n \leq K$

$p(2), p(3), p(4) \ldots p(K) \rightarrow$ true

Now $n = K+1$

a) If $K+1$ is prime, then $p(K+1)$ is true

b) if $K+1$ is not prime ⇒

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$\therefore K+1 = u \times v$ (Factor)

$2 \leq u \leq K, \quad 2 \leq v \leq K$

$P(u), P(v)$ is true

i.e $K+1$ is also a prime number.

⇒ Recursive functions in discrete Mathematics

⇒ Recursion refers to process in which a recursive process repeats itself.

If $\langle a_n \rangle$ be a sequence, an equation connecting a_n with a finite number of other terms as $a_{n-1}, a_{n-2}, \ldots, a_{n-k}$ of the sequence is called recurrence relation.

Example: Fibonacci sequence

$\langle 1, 1, 2, 3, 5, 8, 13, \ldots \rangle$

So $a_n = a_{n-1} + a_{n-2}$ is a reccurence relation.

Note: → This relation is also called difference equation

⇒ ORDER OF RECURRENCE RELATION

⇒ The order of recurrence relation is difference between Greatest suffix & least suffix.

Example:- Let $a_n = a_{n-1} + a_{n-2}$

then order is $n-(n-2) = 2$

⇒ Recursively Defined Function

We use two steps to define a functions with the set of non-negative integers as it domain:

1. **Basis step**:- specify the value of the function at zero.
2. **Recursive Step**:- Give a rule for finding its value at an integer from it's values at smaller integers.

Such defn is called "Recursive step".

Ex:→ Suppose that f is defined recursively by

$f(0) = 3$, $f(n+1) = 2f(n) + 3$

find $f(1)$, $f(2)$, $f(3)$, and $f(4)$

Solution:→

$$f(1) = f(0+1) = 2f(0) + 3 = 2 \times 3 + 3 = 9$$

$$f(2) = f(1+1) = 2f(1) + 3 = 2 \times 9 + 3 = 21$$

$$f(3) = f(2+1) = 2 \times 21 + 3 = 45$$

$$f(4) = f(3+1) = 2 \times 45 + 3 = 93$$

Ex:→ Give a recursive definition of

$$\sum_{k=0}^{n} a_k$$

Soln:→

i) $\sum_{k=0}^{0} a_k = a_0$

ii) $\sum_{k=0}^{n+1} a_k = a_0 + a_1 + a_2 + \cdots a_n + a_{n+1}$

$$\boxed{\sum_{k=0}^{n+1} = \sum_{k=0}^{n} a_k + a_{n+1}}$$

Ex:- Obtain recursive definitions for the function
$f(n) = a_n$ where $a_n = 5n$

Soln:-) i) $f(0) = a_n = 5n = 5\times0 = 0 = a_0$

$f(1) = 5\times1 = 5 = a_1$

$f(2) = 5\times2 = 10 = a_2$

$f(3) = 5\times3 = 15 = a_3$

We can rewrite these as

$a_0 = 0$ and $\boxed{a_n = a_{n-1} + 5}$

for $n \geq 1$

=> Linear Recurrence Relation

=> A recurrence relation having only the first power of the sequence $\langle a_n \rangle$ & not having any product term is said to be linear

Example:-> $5a_n - 3a_{n-1} + a_{n-2} = 0$

-> Linear Recurrence Relation with Constant Co-efficient:

A recurrence relation of the form

$c_0 a_n + c_1 a_{n-1} + \cdots c_k a_{n-k} = f(n)$

is called linear recurrence Relation with Constant Co-efficient.

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Solution of Linear Recurrence Relation with Constant coefficient:-

Working Rule:-

i) Find auxiliary equation

ii) Case-1. When roots are real & distinct then

$$y_n = A_1(a_1)^n + A_2(a_2)^n \cdots A_k(a_k)^n$$

Where $a_1, a_2 \cdots a_k$ are roots

Case-2. When roots are equal then

$$y_n = (A_1 + A_2 n + A_3 n^2 \cdots A_m n^{m-1}) a^n$$

Case-3 When roots are non repeated complex numbers

$\alpha + i\beta$ then $y_n = r^n(A_1 \cos n\theta + A_2 \sin n\theta)$

Where $r = \sqrt{\alpha^2 + \beta^2}$

Example:→ Solve the recurrence relation

(1) $a_{n+2} - 3a_{n+1} + 2a_n = 0$

Solution:→ put $a_n = x^n$

Then $x^{n+2} - 3x^{n+1} + 2x^n = 0$

$\Rightarrow x^n(x^2 + 3x + 2) = 0$

$\Rightarrow x^2 - 3x + 2 = 0$ — Auxilary eqn

$\Rightarrow x^2 - 2x - x + 2 = 0$

$x(x-2) - 1(x-2) = 0$

$(x-1)(x-2) = 0$

$x = 1, 2$

So, solution $a_n = A_1(2)^n + A_2(1)^n = A_2 + A_1(2^n)$.

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ii) $a_n = 4a_{n-1} - 4a_{n-2}$

Solution! -> put $a_n = x^n$

$x^n = 4x^{n-1} - 4x^{n-2}$

$\Rightarrow x^n - 4x^{n-1} + 4x^{n-2} = 0$

$\Rightarrow x^{n-2}(x^2 - 4x + 4) = 0$

$x^2 - 4x + 4 = 0$

$x = 2, 2$

So, solution $a_n = (A_1 + A_2 n) 2^n$

iii) $a_n + 2a_{n-1} + 4a_{n-2} = 0$

Solution! -> put $a_n = x^n$

$x^n + 2x^{n-1} + 4x^{n-2} = 0$

$x^{n-2}(x^2 + 2x + 4) = 0$

$x^2 + 2x + 4 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{4^2 - 16}}{2}$

$= \frac{-2 \pm 2\sqrt{3}\, i}{2}$

$= -1 \pm \sqrt{3}\, i$

$\alpha = -1, \beta = \sqrt{3}$

$r = \sqrt{\alpha^2 + \beta^2} = \sqrt{1+3} = 2$

$\theta = \tan^{-1}(\beta/\alpha) = \tan^{-1}(\sqrt{3}/-1)$

$= \tan^{-1}(-\sqrt{3})$

$= \pi - \pi/3 = 2\pi/3$

So, Solution is

$$a_n = (2)^n \left(A_1 \cos n\, 2\pi/3 + A_2 \sin n\, 2\pi/3\right)$$

⇒ <u>Analysis of Recurrence relation of Binary search</u>

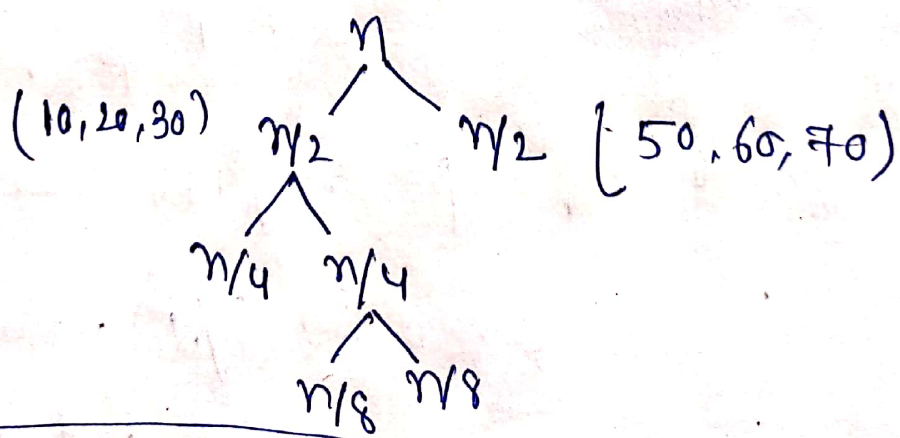
⇒ Binary search (a, i, j, x)

```
{  mid = (i+j)/2
   if (a[mid] == x)
       return (mid);
   else
       if (a[mid] > x)
           Binary search (a, i, mid-1, x)
       else
           Binary search (a, mid+1, j, x)
}
```

Example:- 10, 20, 30, 40, 50, 60, 70

i = 1, 4, j = 7

⇒ mid = $1+7/2 = 4$



$$\boxed{T(n) = T(n/2) + c}$$